

Protocol: In Vitro Tau Aggregation Assay

1. Reagents and Buffer Preparation

- Recombinant Tau (1-441): Purified according to lab standard SOP-03.
- Heparin (H3393): Dissolved in water at a stock concentration of 10 mg/mL.
- Thioflavin T (ThT): Prepared as a 1 mM stock solution, filtered at 0.22 μ m.
- Buffer: 10 mM HEPES, 100 mM NaCl, pH 7.4.

Crucial Note: All reagents must be kept on ice during the assembly.

2. Assay Assembly

Step 1: Dilute Tau protein to a final concentration of 10 micro-molar (μM) in buffer.

Step 2: Add Heparin to the mixture at a 1:1 weight ratio relative to Tau.

Step 3: Add ThT to a final concentration of 20 μM .

Step 4: Pipette 100 μL of the mixture into a black 96-well plate (half-area).

Step 5: Seal the plate and incubate at 37 degrees Celsius with continuous shaking (300 rpm).

Measurement: Record ThT fluorescence every 10 minutes (Ex: 440 nm, Em: 480 nm).